# LLMs for Qualitative Work

## D-Lab AI Pulse Workshop

Materials for all Pulse Workshops available here:
https://github.com/dlab-berkeley/AI_Pulse

Form for suggesting tools/tasks you would like to see covered in future workshops:
https://forms.gle/x8KfejgCViE7t6iQA

# The Bottleneck: Qualitative Research at Scale

The traditional workflow for 40 interviews:

- Transcribe. Then read. Then read again.
- Generate codes, reconcile with a co-coder, write analytic memos.
- Re-read everything through a different theoretical lens.
- Start over when a reviewer asks you to reframe.

A single round of thematic analysis on 30 interviews can take months.

LLMs do not replace this work. But they can change where you spend your time.

# Reshaping, Not Replacing

Wang (2025) — a sociology PhD at Tsinghua — treated his own LLM adoption as an autoethnographic field site.

His core metaphor: Prompt engineering is a new research language.

- Grammar = The methodology you already know
- Vocabulary = Your theoretical insight

Designing prompts forced him to explicitly define his analytical framework. "What is a cognitive frame for AI risk? What are its core dimensions?"

Making it clear to the AI was itself structured theoretical thinking.

Companion reading: Wang (2025), "AI for Social Science." Open access, link in Resources.

# What Can LLMs Actually Do for Qualitative Researchers?

| Task | What the LLM does | Tools |
|---|---|---|
| Document analysis | Semantic search, cross-reference, summarize across a corpus | NotebookLM, Claude Projects |
| Qualitative coding | Suggest codes, classify passages, justify decisions | Gemini, ATLAS.ti AI |
| Visual content | Describe, structure, and compare images or video frames | Gemini, GPT-4o |
| Text extraction | Convert open-ended responses into structured data | Gemini API, spacy-llm |
| Research piloting | Generate synthetic respondents to test instruments | Any LLM |

In every case: you define the framework. The LLM applies it. But how reliably?

# The Hallucination Problem: Why Grounding Matters

Hagar et al. (2025): Tested ChatGPT, Gemini, and NotebookLM on 300 documents about TikTok litigation and policy.

| Tool | Responses with hallucinations | Avg. errors per response |
|---|---|---|
| ChatGPT | ∼40% | 1.5 |
| Gemini | ∼40% | 4.0 |
| NotebookLM (RAG) | ∼13% | 3.0 |

The critical nuance: Most errors were not invented facts — they were **interpretive overconfidence**. Models added unsupported claims and transformed attributed opinions into general statements.

For qualitative work, this is more dangerous than a fake quote: it is subtler and harder to catch.

# Today: Five Modules

1. Grounded Document Analysis (NotebookLM)
   Berkeley's student housing crisis
2. Dialogical Qualitative Coding (Gemini)
   Student food insecurity in higher education
3. Multimodal Analysis: Photos and Video (Gemini)
   Vaping normalization on TikTok
4. Structured Extraction from Text Data (Gemini)
   Campus food access barriers survey
5. Simulating Participants and Pilot Testing (Gemini)
   Campus mental health services

Each module: brief context, live demo, limitations.

# Module 1

## Grounded Document Analysis

with NotebookLM

# Grounded Document Analysis

Traditional workflow: Read → highlight → code → cross-reference. Hard to scale across dozens of documents.

RAG (Retrieval-Augmented Generation): Every response is grounded in your uploaded corpus, with inline citations pointing to the source.

Ideal for:

- Policy analysis and institutional discourse
- Legal and regulatory text comparison
- Literature reviews and archival research

Liu & Sun (2025), AERA Open: GPT-4 + human expertise on K–12 policy interviews — 96% alignment at broad theme level (i.e., human coders and the model agreed on which major themes were present, not on every fine-grained code assignment).

# Demo 1: Housing Crisis in NotebookLM

Research question: How do different stakeholders frame Berkeley's student housing crisis?

Setup: Four documents pre-loaded — Jiang-Brittan (2025), Chancellor's Housing Initiative, Housing Task Force report, Policy Review @ Berkeley analysis.

Demo queries:

1. "How does each document frame the causes of the crisis?"
2. "Where do the university and the city disagree on housing responsibility?"
3. "What solutions does each document propose, and who benefits?"

Explore further: Claude Projects, Gemini with Drive, ChatGPT file analysis

**Live notebook:** notebooklm.google.com (UC Berkeley Student Housing Crisis)

# Module 2

## Dialogical Qualitative Coding

with Gemini

# Dialogical Qualitative Coding

The process is dialogical — a conversation between researcher, data, and LLM — not a one-shot extraction.

Wang (2025): The LLM's greatest value in coding may not be the codes it produces, but how it forces the researcher to be rigorous about their own framework.

Ideal for:

- Thematic analysis (Braun & Clarke) and grounded theory
- Content analysis across large interview sets
- Open-ended survey responses

Morgan (2023): Structured prompting for focus group data. Wachinger et al. (2025): Embedding theoretical perspectives in prompts.

# Demo 2: Food Insecurity Coding in Gemini

Setup: Synthetic interview transcript ("Maya," Berkeley junior discussing food access, shame, and coping strategies).

Four-step dialogical workflow:

1. **Role-set**: "You are a qualitative research assistant expert in thematic analysis."
2. **Extract**: "Find all passages related to Coping Strategies."
3. **Classify**: Apply our 8-code codebook. Ask the LLM to justify decisions.
4. **Iterate**: Challenge a coding. "Isn't the arc here about shame, not support?"

Bonus: Re-code from a critical pedagogy lens. The theory changes what you see.

Explore further: ATLAS.ti, MAXQDA AI Assist, CoLoop

**Live demo:** gemini.google.com/share/bd7a5df2187f

# Module 3

## Multimodal Analysis

Photos and Video with Gemini

# Multimodal Analysis: Photos and Video

Visual and video data sit at the core of ethnography, visual sociology, UX research, and social media studies.

The bottleneck: describing and coding visual material is even more time-consuming than text.

Multimodal LLMs (Gemini, GPT-4o) generate structured first-pass descriptions from images and video keyframes.

Key limitation: Better at concrete observation (objects, settings, activities) than emotional or abstract interpretation. The researcher's expertise fills this gap.

Bartolome & Niu (2024): LLM-assisted video analysis of YouTube shorts on depression — more accurate on objects and activities than emotions or genre.

# Demo 3: Vaping on TikTok in Gemini

Anchor study: Vassey et al. (2024) manually coded 14,002 TikTok posts for setting, aesthetics, social context, and normalization cues.

Demo prompts:

1. Structured observation: setting, people, objects, composition, emotional tone, normalization themes.
2. Re-code from a public health lens: What health information is absent? What normalization techniques are present?
3. Cross-image comparison: What visual strategies do these images share?

Key point: The same image reveals different patterns depending on the analytical frame.

Explore further: GPT-4o vision, Claude with vision, Google Cloud Video Intelligence

**Live demo:** gemini.google.com/share/e3fb3c20a1bf

# Module 4

## Structured Extraction

from Text Data with Gemini

# Structured Extraction from Text Data

Many qualitative researchers have large volumes of open-ended responses or archival text where they need structured extraction: entities, sentiments, categories.

This used to require NLP expertise — custom models, regex pipelines. LLMs now do zero-shot named entity recognition, sentiment analysis, and relationship extraction via natural language.

Wang (2025) calls this the "translator" role: natural language research intent $\rightarrow$ structured output.

Hayes (2025), IJQM: LLMs grasp semantic relationships beyond what word-frequency methods capture.

# Demo 4: Campus Food Access in Gemini

Data: 15 open-ended responses to "What barriers have you faced in accessing food on campus?"

**Version A (no code):**

- Paste responses + extraction schema into Gemini
- Extract per response: Barrier Type, Level (intrapersonal / interpersonal / institutional), Sentiment, Urgency
- Output as a structured table, ready for spreadsheet analysis

**Version B (Python):** 20-line Colab script using the Gemini API. Batch-processes all responses to CSV. Scales to 1,500 responses with no extra effort.

Explore further: spacy-llm, LangChain + Pydantic, Instructor library

**Live demo:** gemini.google.com/share/4487298e7148

# Going Further: LangExtract

Raw API calls work for 15 survey responses. For 300 interview transcripts, you need more infrastructure.

LangExtract (Google, open source) adds four things on top of any LLM:

- **Source grounding**: every extraction maps to an exact character span in the original text — verifiable, not just trusted
- **Schema enforcement**: few-shot examples define your codebook; the output is always consistent
- **Long documents**: automatic chunking, parallel processing, multi-pass for full recall
- **Interactive HTML**: review thousands of extracted entities highlighted in their original context

Works with Gemini (cloud) or Ollama (local — for IRB-protected data).

Companion notebook: `demos/module4_structured_extraction/langextract_demo.ipynb`

# Module 5

## Simulating Participants

### and Pilot Testing with Gemini

# Simulating Participants and Pilot Testing

Wang (2025): "When I first encountered silicon samples, my initial reaction was strong skepticism, even a hint of instinctive resistance."

The argument is not replacing human subjects — it is a virtual testbed for pre-testing hypotheses, piloting instruments, and exploring designs before expensive fieldwork.

Empirical validation:

- Park et al. (2024): 1,052 simulated individuals, 85% accuracy on General Social Survey items
- Hewitt et al. (2025): GPT-4 predictions correlated 0.85 with outcomes of 476 RCTs

Caveat: LLMs flatten variation, narrow responses, and may reinforce stereotypes.

Zhang, Xu & Alvero (2025): Real survey respondents are already using AI to answer open-ended questions, causing data contamination. The issue is already here.

# Demo 5: Mental Health Survey in Gemini

Survey topic: Campus mental health services at UC Berkeley.

Three contrasting personas:

- Jordan (22, CS PhD, Asian American, first-gen, skeptical of therapy)
- Dr. Patricia Okafor (58, tenured professor, Nigerian-American, systemic critic)
- Sam (27, social work MA, non-binary, harm reduction advocate)

Each responds to 5 open-ended questions on accessing mental health support.

Then: Ask the LLM to evaluate its own simulation. Where might it reinforce stereotypes? What did it flatten?

**Live demo:** gemini.google.com/share/dc433993d66c

# Confidential Data and Local LLMs

This cuts across all five modules. Three tiers of data protection:

1. **Commercial cloud** (NotebookLM, Gemini, ChatGPT): Data leaves your machine. Acceptable for de-identified data and public documents. Always check institutional policy.
2. **Berkeley Google Workspace license**: UC Berkeley's campus agreement covers Gemini and NotebookLM. When logged in with a Berkeley account: data is not used to train AI models and is not shared externally. Meaningfully more protective than the consumer versions. (technology.berkeley.edu)
3. **Fully local** (Ollama + OpenWebUI): Open-source models on your laptop. No data leaves your machine, ever. Requires ∼16 GB RAM.

For IRB-protected data with identifiable health information, local deployment may be the only ethical option.

# Using Local LLMs (Inference)

Running a pre-trained model to perform tasks on your data. No GPU required; works on any modern laptop with Ollama.

**Research Examples:**

- **Sensitive Data**: Extracting themes from IRB-protected medical transcripts without data leaving your machine.
- **Pilot Testing**: Quickly generating 100 synthetic responses to a new survey instrument to test for question clarity.
- **Edge Computing**: Running a small model (Llama 3.2 1B) on a tablet during field observations for real-time tagging.

**Recommended Models:**

- **Llama 3.2 (1B/3B)**: Fast, lightweight; runs on any modern laptop.
- **Phi-4 Mini (3.8B)**: Excellent reasoning for its small footprint.
- **Mistral 7B / Gemma 3**: Strong all-around performance.

# When to Fine-Tune Local LLMs

Adapting a model to your specific domain, terminology, or codebook style. Requires GPUs (e.g., Berkeley's Savio HPC).

**Use Cases for Fine-Tuning:**

- **Domain Specificity**: Your research uses highly specialized jargon (e.g., 19th-century legal Latin or specific medical sub-fields).
- **Style Alignment**: You have 500 interviews coded by experts and want the model to mimic that specific, nuanced coding style.
- **Low-Resource Languages**: Improving performance on a dialect or language that is under-represented in base models.

**Berkeley Savio**: Free GPU nodes for researchers. Contact Research IT to get started.

# What We Covered

| | Task | Topic | Tool |
|---|---|---|---|
| 1 | Document Analysis | Berkeley housing crisis | NotebookLM |
| 2 | Qualitative Coding | Student food insecurity | Gemini |
| 3 | Multimodal Analysis | Vaping on TikTok | Gemini |
| 4 | Structured Extraction | Campus food access | Gemini |
| 5 | Simulating Participants | Campus mental health | Gemini |

Through all of these: The researcher defines the framework. The LLM executes. You remain the verifier.

# Tools to Keep an Eye On

| Tool | What it does | Link |
|---|---|---|
| ATLAS.ti AI | AI-assisted coding, auto-code suggestions, and sentiment analysis built into the ATLAS.ti workspace. | atlasti.com |
| MAXQDA AI Assist | Summarize documents, suggest codes, and generate memos without leaving MAXQDA. | maxqda.com |
| CoLoop | Collaborative AI coding platform designed for teams doing qualitative analysis at scale. | coloop.ai |
| AITurk | Run LLM-based simulated surveys and compare responses to human samples. | — |
| spacy-llm | Zero-shot NLP (NER, classification, extraction) via LLMs, built for production pipelines. | spacy.io |

# Discussion

1. **Whose interpretation is it?** If an LLM identifies a theme, is it your finding? What does researcher positionality mean when AI is in the loop?
2. **The deskilling risk**: Is the deeper immersion in data — the craft of reading line by line — being lost? Calculator debate for qualitative methods?
3. **The 13% problem**: Even grounded tools hallucinate 1 in 8 responses, with interpretive overconfidence. How do we build verification habits?
4. **Data sovereignty**: When you upload transcripts to Gemini, who has access? When must you go local?
5. **Silicon samples**: When simulated participants replicate an experiment, what have we learned about human society — and what haven't we?

# Resources

Companion reading: Wang (2025), "AI for Social Science." https://onlinelibrary.wiley.com/doi/10.1002/aaai.70046

Tools used today:

- NotebookLM: https://notebooklm.google.com/
- Gemini (free for Berkeley): https://gemini.google.com/

Demo links:

- Demo 1 (Document Analysis): NotebookLM — UC Berkeley Housing Crisis
- Demo 2 (Coding): Gemini — Food Insecurity Coding
- Demo 3 (Multimodal): Gemini — Vaping on TikTok
- Demo 4 (Extraction): Gemini — Campus Food Access
- Demo 5 (Simulation): Gemini — Mental Health Survey

Local LLMs for sensitive data: Ollama (https://ollama.com/)

D-Lab: https://dlab.berkeley.edu/

All Pulse materials: https://github.com/dlab-berkeley/AI_Pulse